Harsh Raj

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Summary

Final-year B.Tech Computer Science student who builds and fixes production backends in Python and Go: REST APIs, AWS, Docker, CI/CD and release pipelines. **17 merged pull requests** out of 40 across **11 open-source projects** used in production (Kyverno, PipeCD, KEDA, Volcano, Kubeflow, kgateway, Prowler), and reviews of 24 pull requests by other contributors, each finding reproduced with a test.

Education

- **UPES (University of Petroleum and Energy Studies)** *Dehradun, India*
  B.Tech in Computer Science and Engineering (Cybersecurity and Forensics) *2023 – 2027*

Experience

- **Safepay** | Cybersecurity Intern *Jun 2025 – Jul 2025*
  - Triaged **50+ SIEM alerts** by correlating event metadata, affected assets, severity and indicators of compromise.
  - Ran **Nmap** and **OpenVAS** assessments that surfaced 15+ exposed services and misconfigurations.
  - Documented the work as triage playbooks the security team adopted.

Open Source

- **17 merged pull requests, 24 reviewed** | *Go, Python, AWS, Kubernetes, Terraform, gRPC, Docker, GitHub Actions* *GitHub*
  - **Kyverno (6 merged):** made the CLI evaluate namespaced validating policies, which passed in cluster mode while checking nothing (#17608); fixed a panic that took down the mutating webhook on JSON patch values holding object initializers (#17620); fixed double-counted MutatingPolicy results (#17838) and the conformance asserts that kept the E2E gate red for every open PR (#17916). Two of the six are backports.
  - **PipeCD (5 merged), a continuous delivery platform:** fixed **AWS ECS rollbacks** that reported Synced while still running the old revision (#7412); made a cancelled Terraform stage interrupt `terraform apply` so it releases the remote state lock (#7440); three more deployment fixes (#7463, #7477, #7525).
  - **Prowler (1 merged), Python AWS security scanner:** made the EC2 security-group checks flag rules whose CIDRs together cover the whole internet, such as 0.0.0.0/1 plus 128.0.0.0/1 (#13191).
  - **KEDA (2 merged):** fixed a **gRPC connection leak** in the external scaler, measured with the real operator: held connections went 5, 10, 15 over three cycles before the fix and **0** after (#8193, #8213).
  - **Volcano, Kubeflow, kgateway (1 merged each):** fixed a controller crash loop on a misspelled config endpoint (#6057); added API validation so jobs with invalid names are rejected instead of never running (#4155); fixed TLS certificate checks that rejected valid IP and URI certificates (#14760).
  - **Reviews:** 24 PRs reviewed, for example showing that an OpenKruise design's OAuth callback was not bound to the user who started the flow (#1007).

Projects

- **aws-misconfig-scanner – AWS Security Misconfiguration Scanner** | *Python, boto3, AWS, moto, pytest* *2026*
  - Built a read-only scanner in 1,250 lines of Python with **32 checks** across IAM, S3, EC2/VPC, CloudTrail, KMS and RDS, mapped to Security Hub and the CIS AWS Benchmark.
  - Scans every account in an **AWS Organization** through STS role assumption, in parallel; a missing permission reports ERROR, never a silent pass. **55 tests** against moto.
- **sigstore-guard – Image Signature Admission Webhook** | *Go, Kubernetes, Sigstore, Cosign, TUF, PKI* *GitHub*
  - Built a **validating admission webhook** in 2,200 lines of Go that admits only pods whose images are signed by an accepted signer.
  - Anchored trust in **TUF**, so signing-key rotation reaches clusters without a redeploy.
  - Set up CI with an **end-to-end test on kind**.
- **PhishArmor – Multimodal Phishing Detection** | *Python, FastAPI, PyTorch, scikit-learn, Transformers* *GitHub*
  - Built an 8,800-line phishing detector over URL, text and screenshot models, served through a **FastAPI** service with scan, health, activity and dashboard endpoints.
  - Found **37.3%** of query logo crops in its evaluation split pixel-identical to gallery crops; removing them took top-1 accuracy from **0.859 to 0.668**.

Technical Skills

- **Languages:** Python, Go, SQL, Bash, JavaScript
- **Backend and Cloud:** FastAPI, REST APIs, gRPC, Concurrency, AWS (EC2, ECS, S3, IAM, CloudTrail, KMS, RDS), boto3, Docker, Kubernetes, Helm, Terraform, GitHub Actions, CI/CD, Linux
- **Security:** SIEM, Alert Triage, Vulnerability Assessment, Penetration Testing, Nmap, OpenVAS, Burp Suite, OWASP Top 10, IAM (OAuth 2.0, OIDC, JWT), PKI, Policy as Code, Supply Chain Security (Sigstore, Cosign, TUF)
- **Data and AI:** PyTorch, scikit-learn, Hugging Face Transformers, FastAPI, Streamlit, LLMs, RAG
- **Practices:** Unit and Integration Testing, Code Review, Debugging, Root Cause Analysis, Technical Writing